

HOMEWORK 11

INTRODUCTION TO COMPLEX ANALYSIS

(MATH 4230-40), SPRING 2022

SECTION 4.2 - RESIDUE THEOREM

1. (a) Evaluate the contour integral $\int_{\gamma} \frac{e^{-z^2}}{z^2} dz$, where γ is the square with corners $-1-i, 1-i, 1+i, -1+i$.

Note that $\frac{e^{-z^2}}{z^2}$ has a pole of order 2 at the origin, which certainly lies within the region bounded by γ . By Proposition 4.1.6, we have

$$\operatorname{Res}\left(\frac{e^{-z^2}}{z^2}; 0\right) = \frac{d}{dz}(e^{-z^2})|_{z=0} = -2ze^{-z^2}|_{z=0} = 0.$$

By the Residue Theorem, this gives

$$\int_{\gamma} \frac{e^{-z^2}}{z^2} dz = 2\pi i \operatorname{Res}\left(\frac{e^{-z^2}}{z^2}; 0\right) = 0.$$

- (b) Evaluate the contour integral $\int_{\gamma} \frac{e^{-z^2}}{z^2} dz$, where γ is the ellipse $\gamma(t) = a \cos t + ib \sin t$ for $t \in [0, 2\pi]$ and $a, b > 0$.

As the curve $\gamma(t)$ given here is homotopic as a closed curve to the square in part (a), we have

$$\int_{\gamma} \frac{e^{-z^2}}{z^2} dz = 2\pi i \operatorname{Res}\left(\frac{e^{-z^2}}{z^2}; 0\right) = 0.$$

2. Let f be analytic on $\mathbb{C}$ except for poles at 1 and -1 . Assume that $\operatorname{Res}(f; 1) = -\operatorname{Res}(f; -1)$. Let $A = \{z : z \notin [-1, 1]\}$. Show that there is an analytic function h on A such that $h'(z) = f(z)$.

Let f be analytic on $\mathbb{C}$ except for poles at 1 and -1 , and assume that $\operatorname{Res}(f; 1) = -\operatorname{Res}(f; -1)$. Let $A = \{z : z \notin [-1, 1]\}$, and observe that if γ is a closed curve in A , then either no poles or both poles of f lie within the region bounded by γ . That is, $I(\gamma, 1) = I(\gamma, -1)$ for any closed curve γ in A . As such, the Residue Theorem yields that

$$\int_{\gamma} f dz = 2\pi i [I(\gamma, 1)\operatorname{Res}(f, 1) + I(\gamma, -1)\operatorname{Res}(f, -1)] = 2\pi i [I(\gamma, 1)\operatorname{Res}(f, 1) - I(\gamma, 1)\operatorname{Res}(f, 1)] = 0.$$

Correspondingly, since the integral of f around any closed curve in A is zero, it follows from the Path Independence Theorem (i.e. Theorem 2.1.9) that there exists an antiderivative of f on A ; i.e. an analytic function h defined on A such that $h'(z) = f(z)$.

3. (a) Find the residue of $\frac{(z-1)^3}{z(z+2)^3}$ at $z = \infty$.

By definition, the residue of a function $f(z)$ at $z = \infty$ is defined as $\text{Res}(f, \infty) = -\text{Res}((1/z^2)f(1/z); 0)$. As such, writing $f(z) = \frac{(z-1)^3}{z(z+2)^3}$, we need only compute the residue of

$$\frac{f(1/z)}{z^2} = \frac{-(z-1)^3}{z(2z+1)^3}$$

at $z = 0$. Since $z = 0$ is a simple pole of this function, by Proposition 4.1.2 yields that

$$\text{Res}((1/z^2)f(1/z); 0) = \left. \frac{-(z-1)^3}{\frac{d}{dz}(z(2z+1)^3)} \right|_{z=0} = 1.$$

Therefore, we conclude that

$$\text{Res}\left(\frac{(z-1)^3}{z(z+2)^3}; \infty\right) = -1.$$

- (b) Give two methods of evaluating

$$\int_{\gamma} \frac{(z-1)^3}{z(z+2)^3} dz,$$

where γ is the circle with center zero and radius 3.

Notice that the integrand has a simple pole at $z = 0$ and an order 3 pole at $z = -2$, both of which lie within γ . Accordingly, the residue theorem yields that

$$\int_{\gamma} \frac{(z-1)^3}{z(z+2)^3} dz = 2\pi i \left(\text{Res}\left(\frac{(z-1)^3}{z(z+2)^3}; 0\right) + \text{Res}\left(\frac{(z-1)^3}{z(z+2)^3}; -2\right) \right).$$

Using Proposition 4.1.2, we have

$$\text{Res}\left(\frac{(z-1)^3}{z(z+2)^3}; 0\right) = \left. \frac{(z-1)^3}{\frac{d}{dz}(z(z+2)^3)} \right|_{z=0} = -\frac{1}{8}.$$

Using proposition 4.1.6, we have

$$\text{Res}\left(\frac{(z-1)^3}{z(z+2)^3}; -2\right) = \frac{1}{2!} \left(\frac{d^2}{dz^2} \frac{(z-1)^3}{z} \right) \Big|_{z=-2} = \frac{9}{8}.$$

Combining, this gives

$$\int_{\gamma} \frac{(z-1)^3}{z(z+2)^3} dz = 2\pi i \left(-\frac{1}{8} + \frac{9}{8} \right) = 2\pi i.$$

Similarly, using Proposition 4.2.4, we could compute this contour integral as

$$\int_{\gamma} \frac{(z-1)^3}{z(z+2)^3} dz = -2\pi i \text{Res}\left(\frac{(z-1)^3}{z(z+2)^3}; \infty\right).$$

Using our work from part (a), this clearly gives the same result.

SECTION 4.3 - EVALUATION OF DEFINITE INTEGRALS

4. Evaluate $\int_0^\pi \frac{d\theta}{(a+b\cos\theta)^2}$ for $0 < b < a$.

Suppose $0 < b < a$ and write $R(x, y) = \frac{1}{(a+bx)^2}$ and observe that R is a rational function. If $|x| < 1$, then since $b < a$, we have $-bx < b < a$. So, $a+bx \neq 0$, since the contrary would imply that $a = -bx$. Hence, R is a rational function whose denominator does not vanish on the unit circle. Accordingly, by Proposition 4.3.2, the integral $\int_0^{2\pi} R(\cos\theta, \sin\theta)d\theta$ may be computed as $2\pi i$ times the sum of the residues of $f(z) = \frac{1}{iz}R\left(\frac{1}{2}\left(z+\frac{1}{z}\right), \frac{1}{2}\left(z-\frac{1}{z}\right)\right)$ inside the unit circle. We compute:

$$\frac{1}{iz}R\left(\frac{1}{2}\left(z+\frac{1}{z}\right), \frac{1}{2}\left(z-\frac{1}{z}\right)\right) = \frac{1}{iz} \left(\frac{1}{\left(a+\frac{b}{2}\left(z+\frac{1}{z}\right)\right)^2} \right) = \frac{-4iz}{(2az+bz^2+b)^2}.$$

The roots of the denominator above are $z_1 = \frac{-a+\sqrt{a^2-b^2}}{b}$ and $z_2 = \frac{-a-\sqrt{a^2-b^2}}{b}$, both with multiplicity 2, and are correspondingly order 2 poles of $f(z)$. Write $f(z) = \frac{g(z)}{h(z)}$ and note that

$$\begin{aligned} g(z) &= -4iz \\ g'(z) &= -4i \\ h''(z) &= 8a^2 + 4b^2 + 24abz + 12b^2z^2 \\ h'''(z) &= 24ba + 24b^2z \end{aligned}$$

So, we may compute the residues using Proposition 4.1.4 as

$$\text{Res}(f; z_k) = 2 \frac{g'(z_k)}{h''(z_k)} - \frac{2}{3} \left(\frac{g(z_k)h'''(z_k)}{(h''(z_k))^2} \right).$$

Since $a > b > 0$, we have $a^2 > b^2$ and $a^2 - b^2 > 0$. So, both z_1 and z_2 lie on the real axis. Notice that

$$|z_1| < 1, \quad |z_2| > 1.$$

As such, z_1 is contained in the unit circle. By Proposition 4.3.2, we have

$$\int_0^{2\pi} R(\cos\theta, \sin\theta)d\theta = 2\pi i \left[2 \frac{g'(z_1)}{h''(z_1)} - \frac{2}{3} \left(\frac{g(z_1)h'''(z_1)}{(h''(z_1))^2} \right) \right]$$

Since $\frac{1}{(a+b\cos\theta)^2}$ is even, note that

$$\int_0^\pi \frac{d\theta}{(a+b\cos\theta)^2} = \int_0^{-\pi} \frac{d\theta}{(a+b\cos\theta)^2} = - \int_{-\pi}^0 \frac{d\theta}{(a+b\cos\theta)^2}.$$

Thus, we have

$$\begin{aligned} \int_0^{2\pi} \frac{d\theta}{(a+b\cos\theta)^2} &= \int_0^\pi \frac{d\theta}{(a+b\cos\theta)^2} + \int_\pi^{2\pi} \frac{d\theta}{(a+b\cos\theta)^2} \\ &= \int_0^\pi \frac{d\theta}{(a+b\cos\theta)^2} + \int_{-\pi}^0 \frac{d\theta}{(a+b\cos\theta)^2} \\ &= 0 \end{aligned}$$

5. Evaluate $\int_0^\infty \frac{x^{a-1}}{1+x^3} dx$ for $0 < a < 3$.

Write $f(z) = \frac{z^{a-1}}{1+z^3}$ and suppose that $3 > a \geq 1$. Notice that, in this case, f has simple poles at $z_1 = -1$, $z_2 = e^{\pi i/3}$ and $z_3 = e^{-\pi i/3}$. So, the residues are given by

$$\text{Res}(f; z_k) = \left. \frac{z^{a-1}}{3z^2} \right|_{z=z_k} = \frac{1}{cz^{3-a}}.$$

Using Proposition 4.3.11 in the upper half plane, the principal value is

$$\text{P.V.} \int_0^\infty$$

6. Show that $\int_0^\pi \sin^{2n} \theta d\theta = \frac{\pi(2n)!}{(2^n n!)^2}$.